

1 Node.js Introduction

Node.js kunai programming language hoina. Yo JavaScript ko euta **Runtime Environment** हो। [\[1\]](#)

[JavaScript Code] —> [Node.js Runtime] —> [V8 Engine (C++)] —> [Machine Code / OS]

- **V8 Engine:** Google Chrome ko powerful V8 engine lai use garera yesle browser bahira (tapai ko computer ko terminal ma) JavaScript run garchha. [\[1\]](#)
 - **Asynchronous & Non-blocking:** Node.js le euta request ko kaam nasakinjel arko request lai rokerahandaina. Yo single-threaded bhayera pani ekdamai fast chha. [\[1, 2\]](#)
 - **Use Case:** Web servers, APIs, streaming apps (jastai Netflix), ra real-time chat apps banauna use hunchha. [\[1\]](#)
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2 npm (Node Package Manager)

npm duniya ko sabai bhanda thulo software registry ho. Yo Node.js install garda aafai install hunchha. [\[1\]](#)

- **Kaam:** Arule banayera rakhdeko code libraries (packages) haru aafno project ma download garera use garna madat garchha.
 - **Commands:**
 - `npm init` : Naya project setup garna.
 - `npm install <package-name>` : Kunai package download garna (e.g., `npm install express`). [\[1, 2\]](#)
-

3 package.json

Yo tapai ko backend project ko **Kundali/Kundali (Identity Card)** jastai ho. Project ko sabai metadata yahi hunchha.

- **package.json:** Project ko naam, version, ra kun kun third-party libraries use bhairakhya chha (dependencies) jasta kura haru tracking garchha. [\[1, 2\]](#)

- **package-lock.json:** Kun library ko exact kun version download bhako chha ra tyo library chalna arko kun package chahinchha (dependency tree) bhanera detail locker ho. [\[1\]](#), [\[2\]](#)

- **Example Structure:**

```
json
{
  "name": "my-backend-app",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "start": "node index.js"
  },
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

Use code with caution.

4 Modules

Node.js ma harek single JavaScript file euta **Module** ho. Harek file ko aafnai scope hunchha.

- **Core Modules:** Node.js bhitra aafai aune (`fs`, `path`, `os`, `http`).
- **Local Modules:** Tapai aafule aafno project ma banaune file haru (`auth.js`, `database.js`).
- **Third-Party Modules:** `npm` bata download garine package haru (`express`, `mongoose`). [\[1\]](#), [\[2\]](#), [\[3\]](#)

5 require vs import

JavaScript ma euta file ko code arko file ma use garna dui tarika chhan. [\[1\]](#)

Feature	<code>require</code> (CommonJS)	<code>import</code> (ES Modules)
Syntax	<code>const fs = require('fs');</code>	<code>import fs from 'fs';</code>
Standard	Node.js ko purano/default standard.	Modern JavaScript (ES6) standard.

Execution	Synchronous (runtime ma load hunchha).	Asynchronous (compile time mai check hunchha).
Setup	Direct chalthha.	package.json ma "type": "module" thapna parchha.

6 File System (fs Module)

Computer ko hard drive ma file banaune, padhne, update garne, ra delete garne kaam yesle garchha.

- **Example Code (Asynchronous Tarika):**

javascript

```
const fs = require('fs');

// 1. File ma naya data lekhne
fs.writeFile('message.txt', 'Hello Nepal from Node.js!', (err) => {
  if (err) throw err;
  console.log('File successfully baniyo!');

  // 2. Tyo file ko data read garne
  fs.readFile('message.txt', 'utf8', (err, data) => {
    if (err) throw err;
    console.log('File ko data:', data);
  });
});
```

Use code with caution.

[\[1\]](#)

7 Path Module

Harek Operating System (Windows, Mac, Linux) ma file ko location (path) handle garne tarika farak hunchha. path module le tyo cross-platform issue solve garchha.

- **Example Code:**

javascript

```
const path = require('path');
```

```
// Dui saano path haru lai automatic join garera thik path banaucha
const fullPath = path.join(__dirname, 'public', 'images',
'profile.png');
console.log('Main Path:', fullPath);
// Output: C:\your-project\public\images\profile.png (Windows ma)

console.log('File extension:', path.extname(fullPath)); // Output:
.png
Use code with caution.
```

[\[1\]](#)

8 OS Module

Yo module le tapai ko computer (Operating System) ko barema complete details nikalera dinchha.

- **Example Code:**

javascript

```
const os = require('os');

console.log('Platform:', os.platform()); // Output: win32 ba darwin
(Mac)
console.log('Total Memory:', os.totalmem() / (1024 * 1024 * 1024),
'GB');
console.log('Free Memory:', os.freemem() / (1024 * 1024 * 1024),
'GB');
Use code with caution.
```

[\[1\]](#), [\[2\]](#), [\[3\]](#)

9 Environment Variables (.env)

Project ko secret credentials (Database password, API keys, Port numbers) haru code bhitra expose garnu hundaina. Tyaslai .env file ma safe rakhna parchha. [\[1\]](#)

- **Setup Step:** `npm install dotenv` garne.
- **.env File (Secret File):**

env

PORT=5000

DB_PASSWORD=secret_nepal_123

Use code with caution.

- **index.js File (Use garne tarika):**

javascript

```
require('dotenv').config(); // Load variables
```

```
console.log('Server running on port:', process.env.PORT);
```

```
console.log('DB Pass:', process.env.DB_PASSWORD);
```

Use code with caution.

[\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#)

10 Nodemon

Normal Node.js chalaunda code ma euta comma matra change gare pani terminal ma `node index.js` stop garera feri restart garnuparchha. Yo jhyau kaam lai automatic banaune tool **Nodemon** ho. [\[1\]](#)

- **Installation:** `npm install --save-dev nodemon` (Yo devDependency ho, development ko lagi matra chahinchha).
- **Usage:** `package.json` ko scripts ma yesari thapne:

json

```
"scripts": {  
  "dev": "nodemon index.js"  
}
```

Use code with caution.

- **Fayda:** Aba terminal ma `npm run dev` chalaunda, tapaile code save garne bitikkai server automatic restart hunchha! [\[1\]](#), [\[2\]](#), [\[3\]](#)
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